

Short-term prediction of local wind conditions

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Abstract

This paper will describe a system which predicts the expected power output of a number of wind farms. The system is automatic and operates on-line. The paper will quantify the accuracy of the predictions and will also give examples of the performance for specific storm events. An actual implementation of the system will be described and the robustness demonstrated. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

In many areas of wind engineering, predictions of the wind one to two days in advance have become important. These areas include wind energy where electrical utilities with a high wind energy penetration need this information in their scheduling. Transport of large structures as well as other high or low wind-dependent engineering situations spring to mind as other possible applications.

A prediction model has been developed to this end and this paper will describe the model in some detail, but the main focus will be on the performance and operational aspects of the implemented model. The on-line implementation was carried out in an EU JOULE-funded project.

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2. The model

This section will describe the model: the idea behind it and the equations used. Furthermore, an actual on-line implementation will be described.

2.1. The model

The method for predicting the output of a wind farm is outlined in Fig. 1. The idea is to use physical models as ‘far’ as possible, this is done such that the large-scale flow is modeled by a Numerical Weather Prediction (NWP) model, here High Resolution Limited Area Model (HIRLAM) [1] of the Danish Meteorological Institute (DMI); the wind is transformed to the surface using the geostrophic drag law and the logarithmic profile. As we zoom in on the site more and more detail is required; this detail is provided by the Risø WAsP program [2]. WAsP takes local effects (shelter from obstacles, effect of roughness and roughness changes and speed-up/down of hills/valleys) into account. To take the shadowing effects of turbines in a wind farm into account the PARK program [3] of Risø is used. Finally, to take any effects not modeled by the physical model and general errors of the method into account two model output statistics (MOS) modules are used. The model is described and analysed in great detail in [4–6].

2.2. The implementation

To demonstrate that it is possible, not only to obtain accurate predictions of the wind farm production, but also to use these predictions in an on-line environment,

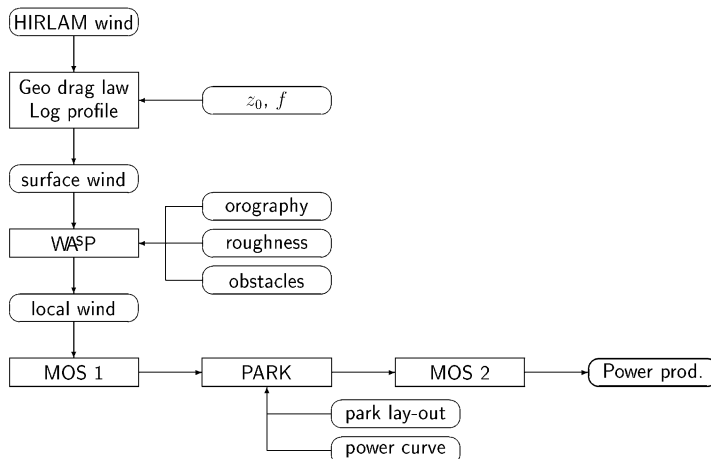


Fig. 1. Flow chart describing the idea behind the model. The prediction from HIRLAM enters at the top and is transformed to the local wind moving down the figure, the production (including shadowing effects from the wind farm) is calculated moving along the bottom line. From [4].

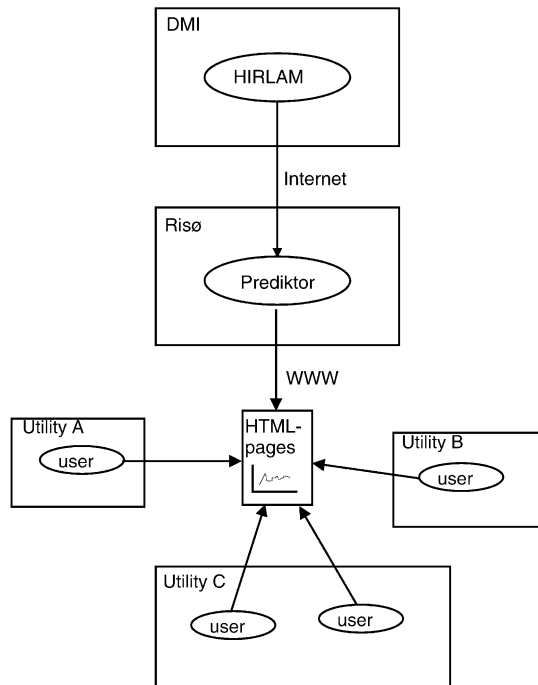


Fig. 2. The on-line implementation. See text for details.

the model was implemented at Risø. The implementation is outlined in Fig. 2. The principle is that as soon as the overall meteorological situation has been predicted by DMI's operational HIRLAM model the forecasts are transmitted via e-mail to Risø. Here a UNIX computer, dedicated to the task, registers the arrival and executes the Risø prediction model, the calculation module then sends the predictions to the output module which generates pages containing the predictions in graphical as well as textual form written in Hypertext Markup Language (HTML). These pages are uploaded to a Web server accessible via any Web browser (as e.g. Netscape and Internet Explorer). The address is not public and the site has no links pointing to it, so only utilities with permission can view the predictions. If the information contained on the pages at some point will be considered as requiring a higher level of security, it is very simple to change the site to be password protected.

Predictions are made for the 47 wind farms shown in Fig. 3 and the model is run twice a day generating predictions 36 h ahead each time.

The model is implemented in such a way that a new wind farm can be added in a matter of a few minutes; HIRLAM predictions for the new site are typically available within 24 h.

Each wind farm can also be changed to represent a region instead of just itself, this is done by scaling the power output with a factor, determined by the total installed capacity of the wind farms in the region.

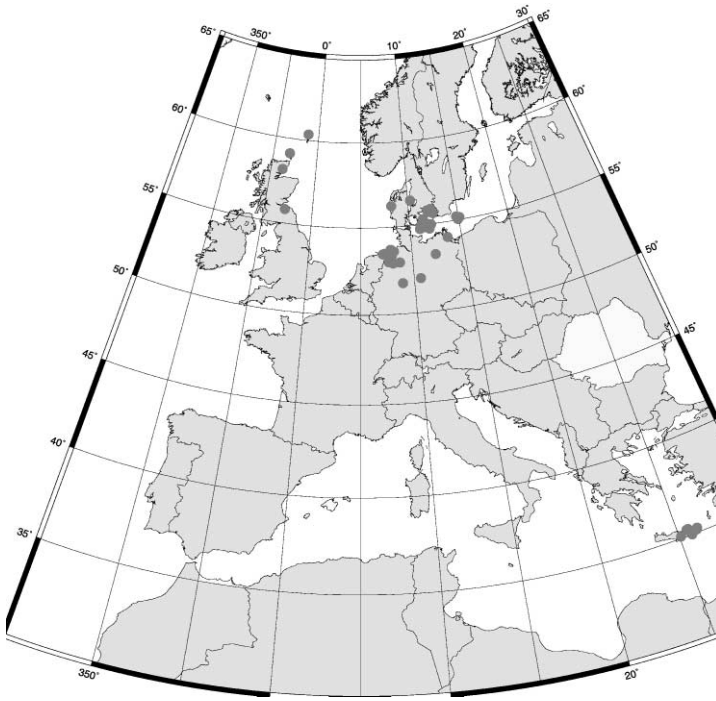


Fig. 3. The location of the 47 wind farms. The wind farms are located in Denmark, UK, Greece and Germany.

The on-line predictions are available to the two Danish utilities, and since the focus of the Risø model is on the Elkraft area, the Elkraft utility is the one which has used the predictions in their daily planning and dispatch. This has been going on for seven months.

3. Results

In this section the quality of the model will be assessed in two ways: the standard way of comparing actual productions to predicted, but also by trying to see how successful the on-line implementation has been. At the end of the section another way of assessing the accuracy and usefulness will be described briefly.

3.1. Accuracy

The model has now run for two entire years and it is possible to draw some firm conclusions as to its performance. For independence of the evaluation, the model is evaluated on one year's worth of data taken from the Elkraft SCADA system. The wind farm observations are recorded every hour.

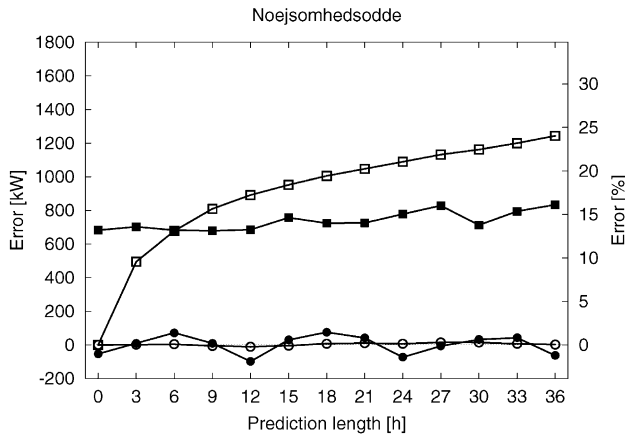


Fig. 4. The mean error (circles) and the mean absolute error (squares) for the prediction of the power output of the Nøjsomhedsodde Wind Farm for the prediction model (filled symbols) and the persistence model (open symbols). The left-hand y-axis shows the absolute numbers (in kW) and the right-hand axis the numbers relative to the total installed capacity (in %). One year's worth of data have been used in the comparison. The farm is rated at 5.2 MW.

To get an estimate of the skill of the model the predictions are compared to those of the *persistence* model, which is a very simple model stating that

$$P(t + \ell) = P(t), \quad (1)$$

where $P(t)$ is the production at time t and ℓ is the look-ahead time. This model could popularly be called the 'what-you-see-is-what-you-get' (WYSIWYG) model! Despite its seeming simplicity, this model describes the flow in the atmosphere rather well, due to the characteristic time scales of weather systems; it is often experienced that the weather in the afternoon is the same as it was in the morning, so for some typical weather situations it is a rather difficult model to beat.

The results of the comparison between the two models for the Nøjsomhedsodde Wind Farm are shown in Fig. 4 where the mean error and the absolute mean error ($= |\text{predicted} - \text{observed}|$, a measure of the scatter, giving less weight to outliers than the standard deviation) of the prediction model are compared to those of the persistence model. The following can be seen:

- The prediction model outperforms the persistence model after 6 h.
- The mean absolute error of the prediction model is around 15% of the installed capacity.
- The decay of the performance of the prediction model is very gentle, over the 36 h time-span the error is only increased by less than 10%.
- The mean error of the persistence model is very small. This follows from the definition of the persistence model.

Table 1

The storms identified in the period 1 Jan 97 to 1 Aug 98. In the column marked 'Quality' the predictions are graded according to the following: – bad, = OK, + good

Start	End	Quality	Notes
1997			
7/1	9/1	–	Level missed
12/1	15/1	–	Start-up missed
18/3	20/3	=	End missed
21/4	24/4	–	Missed completely
10/5	12/5	–	Missed completely
17/11	22/11	+	
25/12	27/12	+	
1998			
14/1	18/1	+	
26/1	29/1	+	
15/2	18/2	=	
25/2	2/3	+	
6/3	8/3	=	
10/3	13/3	–	Level missed
19/3	20/3	=	
15/6	19/6	+	
14/7	16/7	+	

These predictions are not the best and not the worst examples of the performance; a well-predicted wind farm has a scatter as low as 10% of the installed capacity and a not-so-well-predicted one up to 20%.

One very important feature which cannot be seen from this figure is how well the model predicts storms. Here the persistence model has no chance, since it assumes the present status. In a previous study [7] all storms in 1997 and 1998 in the period 1 Jan 97 to 1 Aug 98 were identified for the Avedøre Wind Farm and the performance of the prediction model was evaluated in qualitative terms. The result can be found in Table 1.

The definition of 'storm' is here taken to mean an event where the production rose from a very low value (i.e., close to 0 kW) to around 3000 kW (approx. 80% of the total installed capacity) and then down again to a low value over a period of a few days. Using this selection criteria 7 storms were found in 1997 and 9 in the first seven months of 1998.

These 16 storms are depicted as "thumb-nail images" in Figs. 5 and 6 and listed in Table 1.

Studying the data in detail reveals that of the seven storms in 97 two were predicted very well and one was predicted satisfactorily. In 98 five were predicted very well and three satisfactorily. This is very much in line with the results found for the wind speed at the Risø mast: in the first half of 1997 the HIRLAM model had problems, which then seemed to have been fixed.

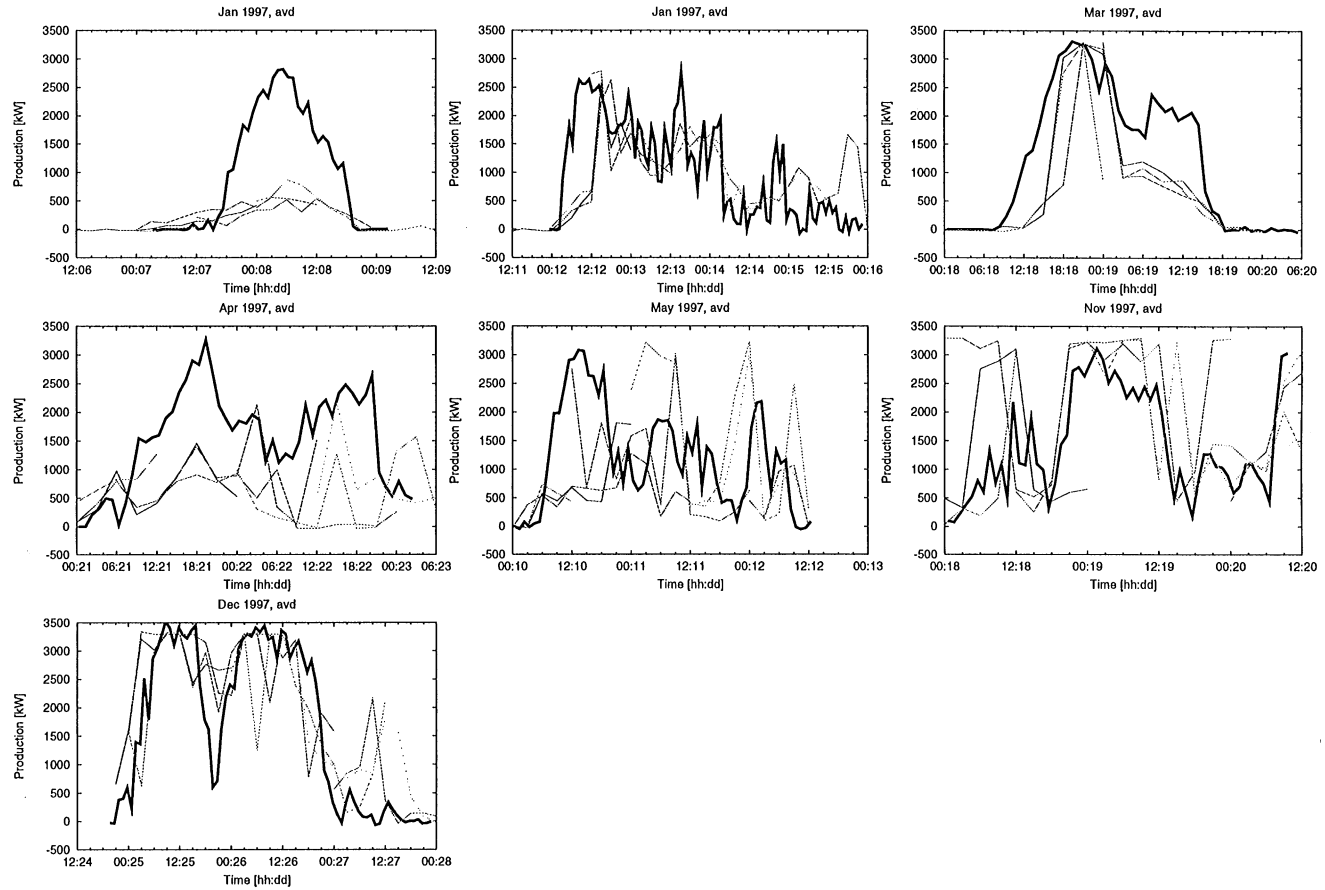


Fig. 5. The storms in 1997 as registered at the Avedøre Wind Farm (capacity 3.6 MW). Solid line is the observed production, dashed lines are the predictions.

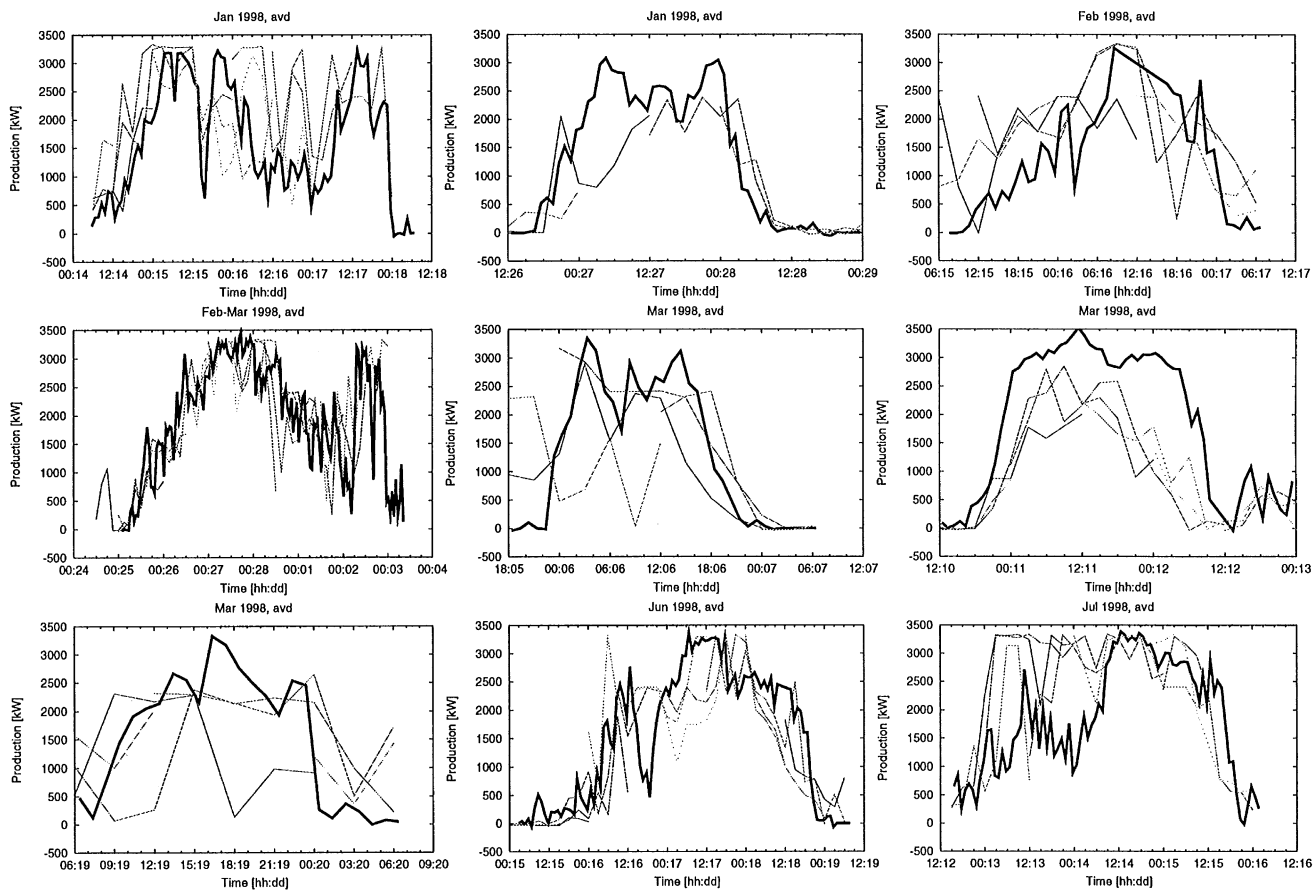


Fig. 6. Storms in 1998 (until 1 Aug) as registered at the Avedøre Wind Farm (capacity 3.6 MW). Solid line is the observed production, dashed lines are the predictions.

The results shown here differ slightly from the results in [7], in that the cases where cut-out (i.e., the wind is so high that the turbine must stop in order to avoid structural damage) was originally predicted are now removed, since the model has been changed to reflect the fact that cut-out very seldomly is seen in the observations. It is critical to get this right, of course, since missing a cut-out situation will make an enormous difference for the dispatch, by either predicting 100% production (no cut-out) or 0% prediction (cut-out).

3.2. Track record/reliability

This section will focus on some operational aspects of the implementation.

The results from the HIRLAM model are typically available after 2 h, the transit time between DMI and Risø is typically less than a minute and the prediction model takes less than 2 min to calculate the predictions and another 2 min to generate the web pages, the upload is instant. So, typically around 2 h after the measurements have been taken at a large number of places around the Northern Hemisphere the predictions are available to the utility.

The system has run operationally for more than 2 yr, with no breakdowns. One planned shut-down took place, when the operating system was upgraded. The only problems have occurred when the DMI forecasts have been delayed for some reason. It is of course not possible to do anything about this from the utility side, so the times different from 2 h are an indication of how often this occurs.

Fig. 7 shows the distribution of the delay time; as can be seen, most of the time the delay is around 2 h and after 4 h 90% of the predictions have arrived. There are some outliers, and as mentioned earlier all of them are due to the fact that the HIRLAM predictions were delayed.

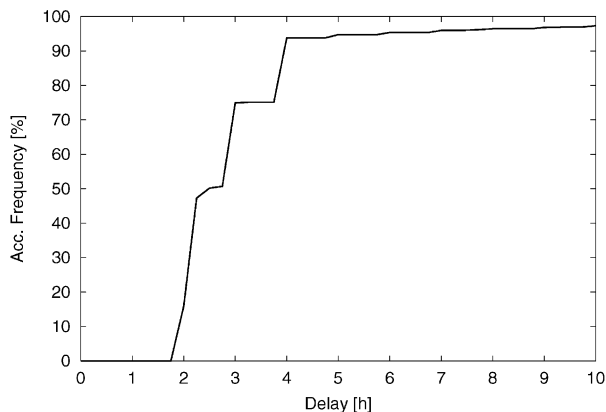


Fig. 7. Accumulated distribution of the delay time of the predictions from the E (previously called DKV) version of the HIRLAM model.

3.3. Savings in fossil fuel

Another very important way of accessing the accuracy from a utility point of view is to see how much conventional fuel and thereby money is saved because of the use of the predictions; other papers are dedicated to this, see [8,9]. The main conclusion is that significant savings are indeed found, the higher the penetration the higher the savings. Typical results are that the use of predictions becomes significant at 25% penetration (note that many areas today have a penetration of more than 20%) and that the savings are more than 25% higher than just using persistence.

4. The future

At present the model is used as-is in a JOULE project, predicting the power for 13 wind farms in Germany (location of the wind farms is shown in Fig. 3).

For the very near future two model refinements are planned, the first one is a direct improvement of the existing model (described in the following) and the second is a completely new model, where the Risø model described here is combined with the IMM model [10]; this development will be carried out in a project funded by the Danish Energy Agency, with the goal of implementing this model at all the Danish electrical utilities.

It is also hoped that other utilities will be interested; Electric Power Research Institute (EPRI) in the US has already shown interest.

4.1. Using the 10 m wind as input

Due to the decrease in grid size and the improvement of the boundary-layer parameterisation of HIRLAM, the level from which the wind has to be taken has changed from that when the model was first developed and implemented (see [11] for further details). In the original model the wind found in model level 5 (approx. 550 m agl) should be used as the best approximation of the geostrophic wind; now, however, the wind from the lowest level has the smallest scatter when compared to the observed wind at the wind farm site. This means that since the original model had to transform the wind from a geostrophic wind via the geostrophic drag law to the surface, the model must be changed, since this transformation has already been done by the HIRLAM model. Referring to Fig. 1 the box labeled ‘Geo. drag law, Log profile’ must therefore be removed. Calculating the mean error and std. dev. of the original model and the model proposed here applied to the Risø mast, the same qualitative picture is found as in Fig. 4; however, the predictions using the 10 m wind show a reduction in the standard deviation of typically 20% on the prediction of the wind speed.

5. Conclusion

This paper has described an automatic on-line prediction system for wind farm production output. The model was described and analysed and its performance was

shown to be good. The on-line implementation was also described and it was shown that the model had been operational for 2 yr with very few interruptions. The predictions have been used by the Danish utility Elkraft for seven months. Lastly, new applications of the model were described.

Acknowledgements

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